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# Magnetization history of the Upper Silurian Dingle Group, SW Ireland

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## SUMMARY

The Dingle Group, a continental Upper Silurian redbed succession in SW Ireland, has yielded a multicomponent palaeomagnetic record. The oldest magnetization retrieved, the **B** component, with *in situ* mean direction of $D/I = 191/+9$; $K = 40$; $\alpha_{95} = 3.6°$; $N = 41$, post-dates the local late Silurian–mid-Devonian (Acadian) tectonic deformation and is in very good agreement with the prevailing remanences in other Caledonian rocks of western Ireland. An overprinted '**A**' component with *in situ* mean direction $D/I = 194/-16$; $K = 53$; $\alpha_{95} = 3.1°$; $N = 41$, is about as prevalent as the **B** component and was probably acquired in late Palaeozoic (Kiaman) times. By comparison with other published data, including a positive fold test for corresponding magnetization of Upper Silurian redbeds in Clare Island, it is concluded that the flat-lying **B** field (relative to Britain and Ireland) was in existence at least from the late Silurian to late Middle Carboniferous, i.e. covering a minimum time span of 100 M.yr.

**Key words:** Ireland, Middle–Upper Palaeozoic, multicomponent palaeomagnetism.

## 1 STUDY PERSPECTIVE

The rock units exposed in the Dingle Peninsula, County Kerry, SW Ireland, range from the marine Wenlock–Ludlow Dunquin Group to the lowest marine Tournaisian (Lower Carboniferous) (Fig. 1). Following the deposition of the fossiliferous marine Silurian, intercalated with acidic volcanics, crustal movement caused local regression and topographic changes, and a broad inland valley formed in the area. In this basin an approximately 4 km thick succession of sandstones, conglomerates and mudstones was laid down in fans of scree, in rivers, and in temporary lakes during Upper Silurian and Devonian times. Despite the existence of many good rock exposures throughout the Peninsula there is a lack of continuity of outcrop, and because of lateral facies changes and the absence of marker horizons and macrofauna, the stratigraphy of the continental redbed succession has long been a matter of debate (*cf.* Horne 1974). The sandstone sequence has been divided into the Dingle, Caherbla and Glengarriff Harbour groups (in ascending order), but further assessments of 'absolute' ages (within the 'Old Red Sandstone' time span) have been hampered by the lack of fossil evidence. Furthermore, these groups of continental sandstone are separated by angular unconformities. This is generally also the situation for the contact between the Dunquin and Dingle groups, but Holland (1987) presents evidence from some inland exposures suggesting stratigraphical continuity from the top beds of the Dunquin Group into local purple siltstones of the overlying Dingle Group. Following Holland (1981) the youngest Silurian fossils found are not 'younger than middle Ludlow in age and thus the Dingle Group includes upper Ludlow strata and, in view of its thickness, most if not all of the Downtonian'. In this view the Dingle Group would be of Upper Silurian age, while the Caherbla and Glengarriff Harbour groups would be ascribed to the Old Red Sandstone proper.

Structurally the Dingle Peninsula is controlled by six major and several minor folds, all striking E–ENE and plunging easterly at angles averaging 10°–15° (Shackleton 1940; Horne 1974). According to Shackleton (1940), the fold system has been progressively tightened during successive tectonic 'pulses' affecting the area, of which the main structural disturbance resulted from two superimposed phases of Caledonian activity. The intensity of these Caledonian crustal movements is well displayed in the famous Bull's Head locality where gently dipping red sandstones of the Glengarriff Harbour Group (Middle–Upper Devonian) rest unconformably on the steeply dipping, flaggy sandstones of the basal Bull's Head Formation of the Dingle Group (Upper Silurian).

The Caledonian deformation was subsequently affected by

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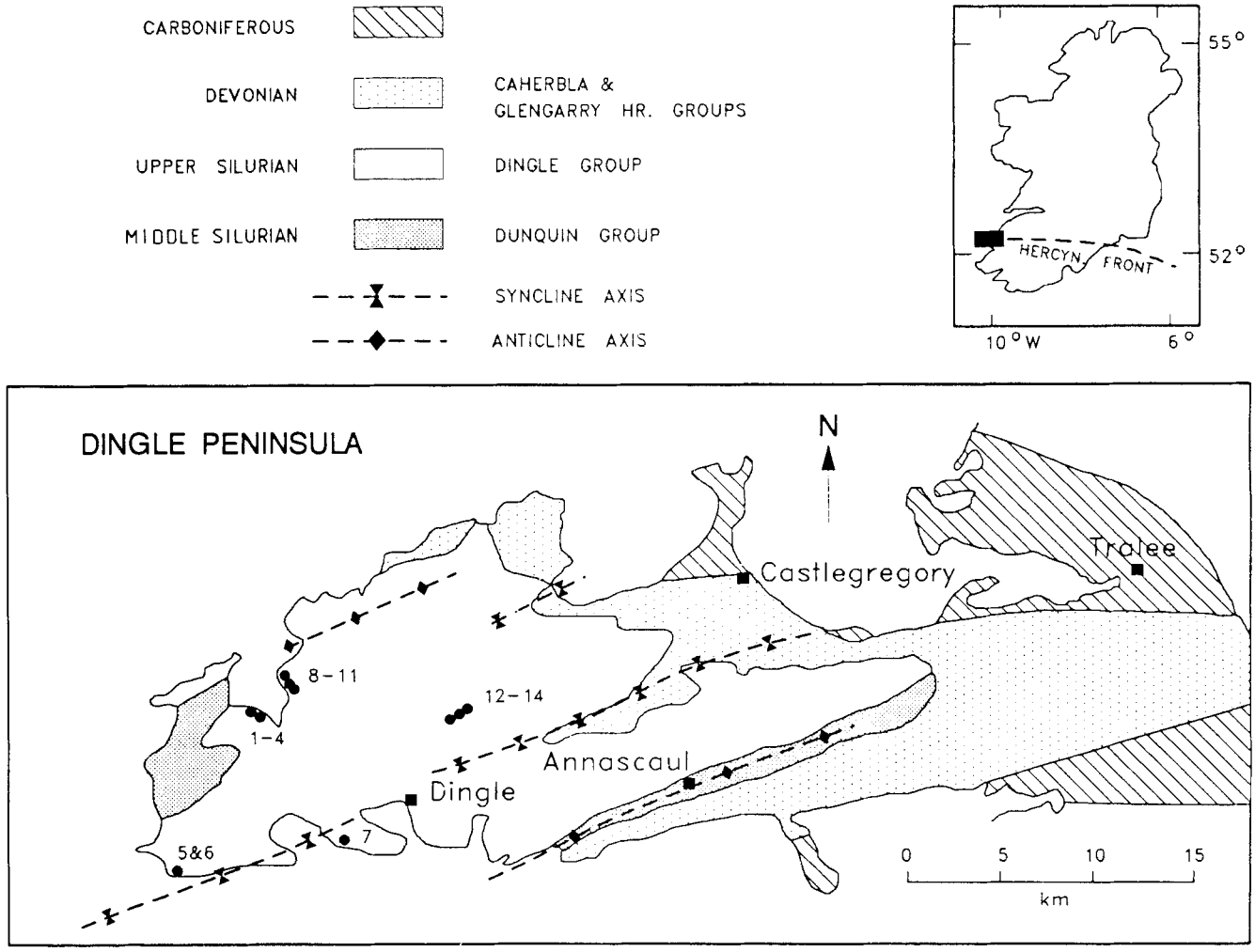



**Figure 1.** Geology of the Dingle Peninsula, simplified from Horne (1974). Some of the major fold axes are depicted to show the general tectonic trend of the region. Numbered circles denote palaeomagnetic sampling sites in the Upper Silurian Dingle Group. For stratigraphic level of sites, see Table 1.

a third, Hercynian phase. The Hercynian tectonics of the Dingle Peninsula form part of the northern rim of a larger external, Rheno-Hercynian, belt which extends from southern Ireland, through southern England, and into Central Europe (see for example, Autran, Breton & Chantraine 1980). In the Dingle Peninsula the Hercynian structures are particularly expressed near the western tip of the Peninsula where they form steeply inclined faults and locally developed cleavage notably in the Dunquin Group (*cf.* Phillips & Holland 1981). However, the overall Hercynian deformation is only minor compared with that associated with the Caledonian tectonic events (Shackleton 1940). This fact is very well demonstrated in the Bull's Head location near Dingle Town (Fig. 1).

Given this background the palaeomagnetic record of the Siluro–Devonian redbed succession of the Dingle Peninsula would be expected to be multicomponent, holding prospective information for studies of Caledonian palaeomagnetism as well as for questions of Hercynian remagnetization. In a recent contribution (Deutsch, Murthy & Storetvedt 1993), we outlined the polyphase remanence record of the Dunquin Group (mid-Silurian) and the Old Red Sandstone sequence (Caherbla and Glengarriff Harbour groups), along with a discussion of certain tectonic aspects of the British and Irish Caledonides. In the present paper, we add further to the palaeomagnetic data base of the Dingle Peninsula, reporting on the magnetization build-up of the Upper Silurian Dingle Group.

## 2 ROCK COLLECTION AND METHODS

The conglomerate–sandstone–shale succession of the Dingle Group has been subdivided into five formations (Horne 1974). According to Horne the basal fine-grained Bull's Head Formation is followed by the mostly thick-bedded, compact and sand-dominated Eask Formation; in the western region these two formations are replaced laterally by the Smerwick Formation, a sandstone–conglomerate succession. Then follows the Trabeg Conglomerate, and finally the sand-dominated top member, the Ballymore Formation.

The palaeomagnetic samples come from fine- to medium-grained sandstone sections of the Eask and Ballymore formations, and from one site of sandstone within the Trabeg Conglomerate. The areal distribution of sites is

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**Table 1.** Sampling details.

| Site | Location* | Formation | Sample nos(Di) | Bedding orientation Strike/dip |
|---|---|---|---|---|
| 1 | Smerwick Harbour South | Ballymore | 1-3 | 020/19ESE |
| 2 | Smerwick Harbour South | Ballymore | 4 & 5 | 020/19ESE |
| 3 | Smerwick Harbour South | Ballymore | 6 | 020/19ESE |
| 4 | Smerwick Harbour South | Ballymore | 7 & 8 | 291/11NNE |
| 5 | Slea Head | Ballymore | 9 & 10 | 065/49SSE |
| 6 | Slea Head | Ballymore | 11 & 12 | 065/49SSE |
| 7 | Ventry Harbour | Trabeg | 13-15 | 241/85NNW |
| 8 | Smerwick Harbour East | Ballymore | 16-18 | 049/32SE |
| 9 | Smerwick Harbour East | Ballymore | 19-21 | 039/30SE |
| 10 | Smerwick Harbour East | Ballymore | 22-24 | 053/35SE |
| 11 | Smerwick Harbour East | Ballymore | 25 & 26 | 034/42SE |
| 12 | North of Connor Pass | Eask | 27-29 | 092/39S |
| 13 | North of Connor Pass | Eask | 30 | 092/39S |
| 14 | North of Connor Pass | Eask | 31-34 | 106/37S |

* See also Fig. 1

depicted in Fig. 1, and further sampling details are listed in Table 1. Orientation of block samples was done with a magnetic compass and dip measurement. All magnetic orientations have been corrected for a local declination of 11°W. Standard cylindrical specimens 1.9 cm in diameter were cut from the 34 block samples collected, and a total of 89 specimens were subjected to thermal demagnetization in steps up to 670° or 680 °C. This experimental work was partly carried out at Bergen University employing a Schonstedt TSM-1 demagnetizer and a Digico magnetometer, and partly at Memorial University where a TSM-1 demagnetizer and a Schonstedt SSM-1 spinner magnetometer were used. Results from individual specimens were first displayed in stereo (or equal area) plots to evaluate their potential for further analytical study by vector methods.

During progressive demagnetization most multicomponent specimens define directional trends. The prerequisite for a meaningful vector analysis of such remagnetization patterns is the existence of a window in the stability spectra for which only one of the components is being erased. Frequently, however, the remagnetization trends are more strongly curved than great circles suggesting a tight interplay of more than two remanence vectors. Such cases are not always properly identified by orthogonal vector diagrams but they are exposed more clearly in stereo or equal area plots. Therefore, a combination of directional diagrams and orthogonal vector plots is necessary in order to reduce potential artificiality in the palaeomagnetic analysis.

## 3 LABORATORY RESULTS

The directions of total natural remanent magnetization (NRM) for all 89 specimens studied are displayed in Fig. 2. To a first approximation the overall directional pattern is suggestive of at least one shallow-inclined, reverse magnetization, plus a subordinate component of normal polarity. Such a palaeomagnetic build-up has frequently been encountered in British Old Red Sandstone rocks (e.g. Storetvedt *et al.* 1990). As we show below the analysis of results of thermal demagnetization has indeed confirmed such a remanence structure.

The most common magnetization found in rocks of the British and Irish Caledonides is the so-called **B** magnetization, according to terminology adopted for the assumed oldest component in Scottish Old Red Sandstone rocks (e.g. Storetvedt & Carmichael 1979; Storetvedt *et al.* 1990). The **B** remanence has shallow inclination (≈10°); it is characterized by a dual-polarity structure but the reverse component normally predominates. Therefore, the superposition of normal and reverse **B** remanences may define resultant NRM directions in intermediate positions but with a strong southerly bias, consistent with the directional pattern of Fig. 2. However, a second magnetization, termed the *A* component, with southerly declination and shallow upward (approximately −20°) inclination, has been identified in a number of British mid-Palaeozoic rocks. Such a system of coexisting **A** and **B** remanences also characterizes the Dingle Group.

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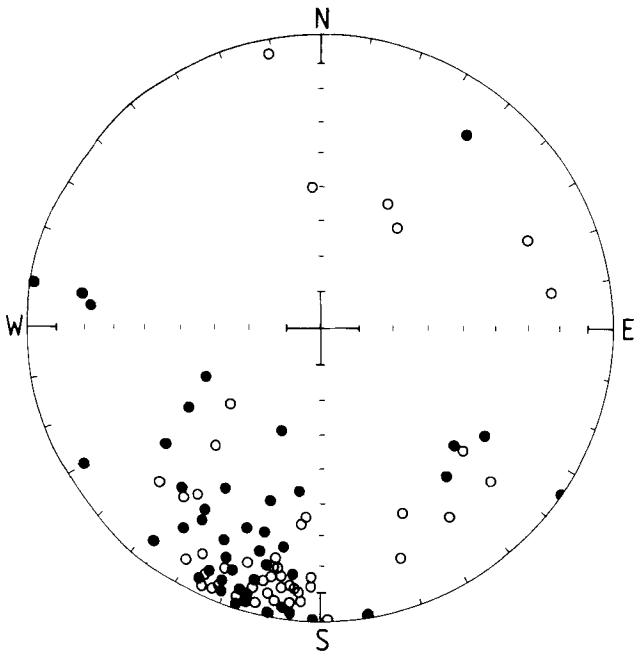



**Figure 2.** Directions of natural remanent magnetization (NRM) for all specimens that have been subjected to demagnetization studies. The diagram is equal area and closed (open) symbols represent downward (upward) inclination.

All palaeomagnetic directions defined in the present study are based on least-square analysis of linear trajectories as seen in orthogonal vector plots. The overall quality of the data can be judged from several examples presented in *in situ* coordinates, in Figs 3–5, depicting a variety of vectorial combinations.

Figure 3 displays data from specimens that are dominated by the **A**-component. Disregarding a minor, soft and scattered remanence with low unblocking temperatures, the successive specimen vectors (on progressive demagnetization), define northerly trending great circle paths (*cf.* Figs 3a and c) during which the shallow southerly, upward-pointing **A**-magnetization is removed. All northerly swinging remagnetization paths in this study are disrupted by experimental noise at various temperatures above 600°C (this stage is disregarded and not shown in the diagrams), i.e. before having reached the terminal, high-temperature (end point) direction. Therefore, the results of specimens that define great circle paths display linear trajectories that bypass the origin, when viewed in orthogonal vector plots. However, when extended, northward swinging remagnetization circles suggest intersection in a shallow position almost due north.

Figures 4 and 5 show examples of **A** and **B** magnetizations. Where the two remanences co-exist (Figs 5c and f), the **B** remanence has the higher of the two blocking temperature ranges.

Of the 89 specimens subjected to thermal treatment only 14 were unsuccessful with respect to component extraction (**A** or **B** or both). Thirty-four specimens gave single **A** directions, while another 34 specimens gave single **B** directions. An additional seven specimens disclosed both magnetizations in co-existence. In total this resulted in 41 directions for each of the two magnetization groups. The two characteristic magnetizations for the Dingle Group are presented in Fig. 6 in *in situ* coordinates. The two

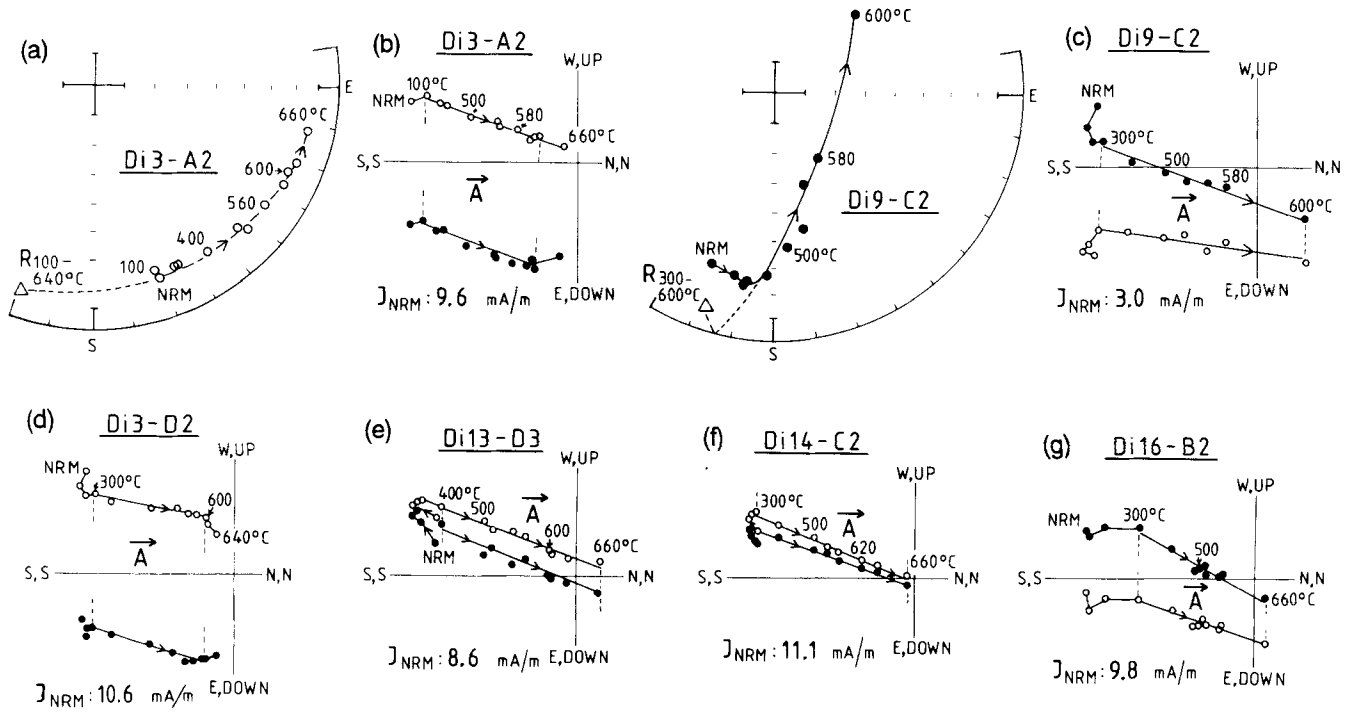



**Figure 3.** Examples of specimens in which the **A** magnetization is the characteristic dominating remanence component. Diagrams (a) and (c) are equal-area plots while the remainders are orthogonal vector projections. Note the general northward progression of the total remanence vector versus increasing demagnetization, indicating the presence of a weak, high-temperature normal magnetization. Triangles [in (a) and (c)] denote directions of erased vectors for temperature range specified. In the vector plots solid (open) symbols are points in the horizontal (vertical) plane. Other conventions are as for Fig. 2.

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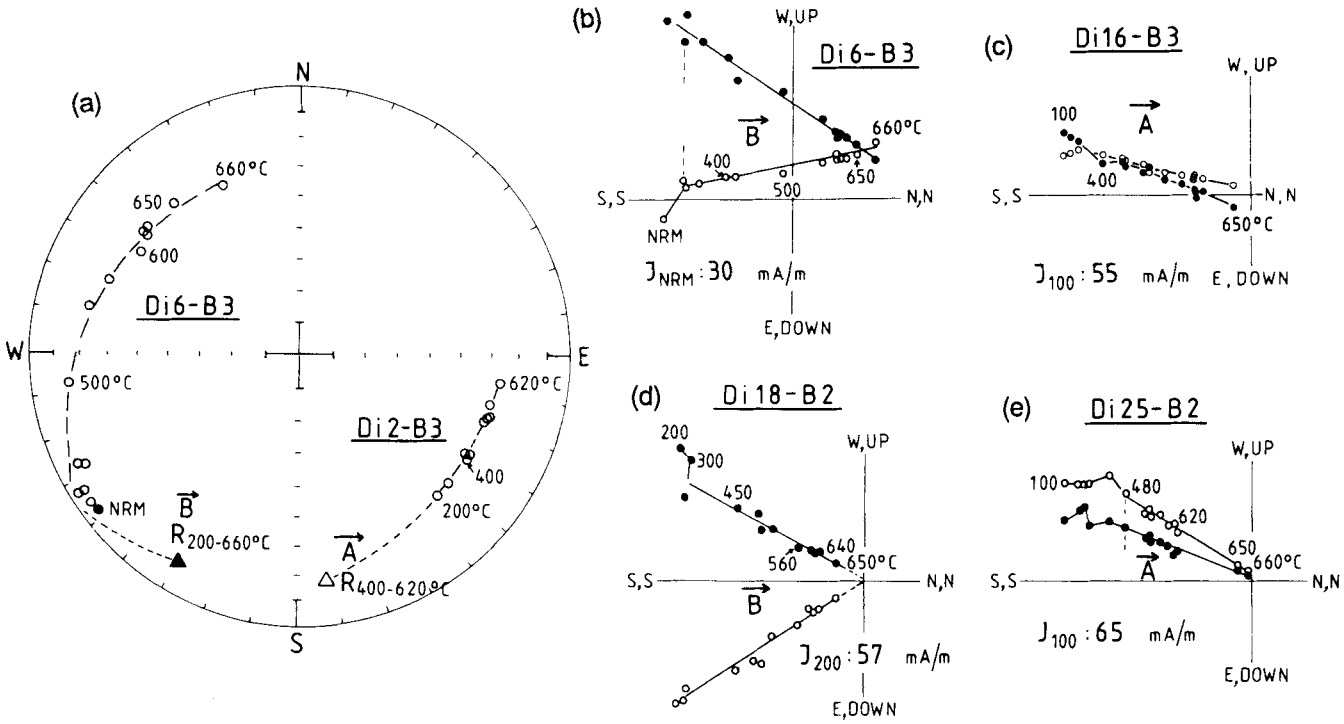



**Figure 4.** Demagnetization results for **A** and **B** specimens. Symbols and conventions are as for Figs 2 and 3.

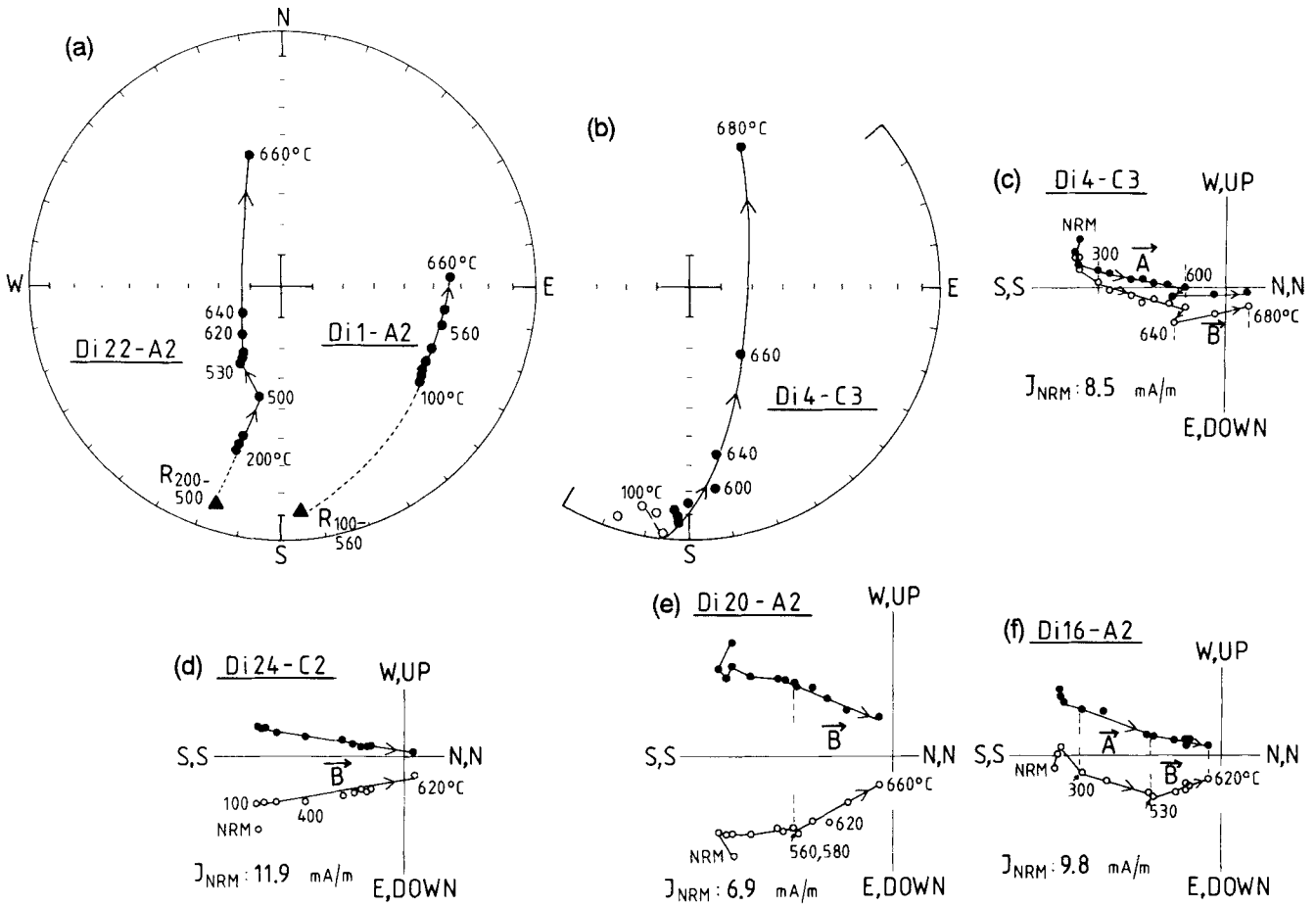



**Figure 5.** Equal-area plots and vector displays of examples for which the **B** component (reverse) plays an important role. Note that diagrams (c) and (f) show components **A** and **B** in coexistence. Again the presence of a high-temperature normal magnetization is indicated by the results. Symbols and conventions are as for Figs 2 and 3.

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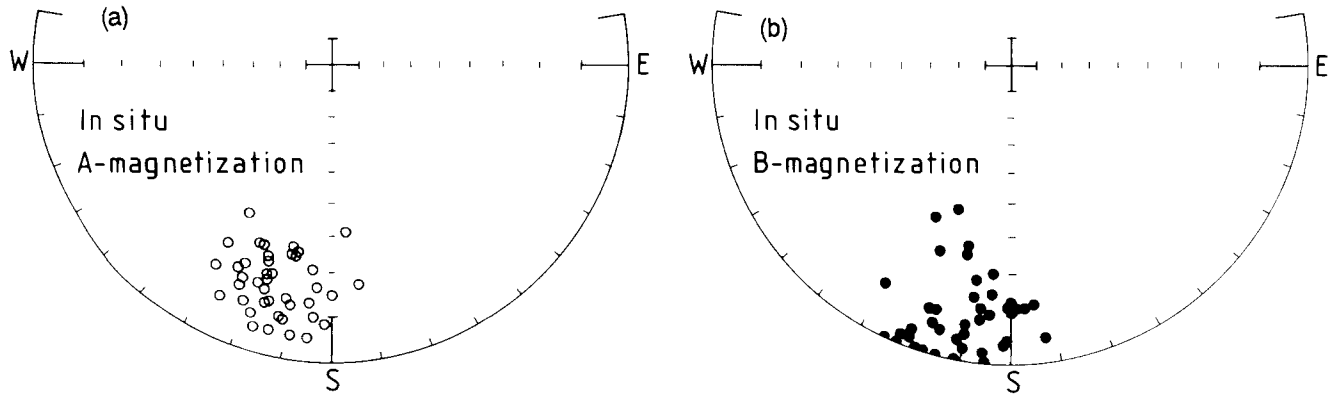



**Figure 6.** Stereographic projections depicting the distributions of **A** and **B** magnetization at specimen level and *in situ* coordinates.

**Table 2.** *In situ* site mean directions of magnetization components **A** and **B** for the Dingle Group, SW Ireland.

| | | Component A | | | | | Component B | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Site | S | N | D | I | k | $\alpha_{95}$ | N | D | I | k | $\alpha_{95}$ |
| 1 | 8 | 2 | 184.5 | -14.5 | | | 4 | 181.5 | 7.5 | 70 | 11 |
| 2 | 9 | 8 | 192.8 | -14.0 | 54 | 7.6 | 3 | 192.5 | 6 | 38 | 20 |
| 3 | 3 | 1 | 206 | -9 | | | 2 | 202.5 | 5 | | |
| 4 | 5 | 3 | 195 | -13 | 66 | 15 | | | | | |
| 5 | 5 | 3 | 190 | -9.5 | 97 | 13 | 1 | 191 | 0 | | |
| 6 | 5 | 2 | 196 | -9.5 | | | 2 | 198 | 7.5 | | |
| 7 | 7 | 6 | 196.3 | -18.5 | 99 | 6.8 | 1 | 200 | 36 | | |
| 8 | 10 | 6 | 201.9 | -22.3 | 85 | 7.3 | 8 | 191.2 | 9.2 | 27 | 10.9 |
| 9 | 4 | | | | | | 3 | 189 | 12.5 | 35 | 21 |
| 10 | 7 | 2 | 190 | -12 | | | 6 | 191 | 11.6 | 51 | 9.5 |
| 11 | 5 | 5 | 186.5 | -20.6 | 42 | 11.9 | 1 | 187 | 9 | | |
| 12 | 8 | 2 | 201 | -12 | | | 2 | 195 | 11 | | |
| 13 | 2 | | | | | | 2 | 195.5 | 1.5 | | |
| 14 | 11 | 1 | 198 | -22 | | | 6 | 188.0 | 9.2 | 48 | 9.8 |

$S$ = number of specimens measured;
$N$ = number of specimen vectors; $D$ = mean declination; $I$ = mean inclination; $k$ = Fisher's precision estimate; $\alpha_{95}$ = radius of the circle of 95 per cent confidence.

populations are well grouped, and they are clearly distinguishable by direction. The palaeomagnetic data at site level are listed in Table 2.

## 4 INTERPRETATION

According to the above, the Dingle Group, of assumed Upper Silurian age, records two superimposed palaeomagnetic field components, **A** and **B**. The **B** remanence is widespread in the British and Irish Caledonides, while the **A** magnetization, though frequently encountered, has a much more patchy occurrence within the orogenic belt. Locally, however, the **A** component is strongly developed as is the case in the Orcadian basin of North Scotland where it has been related to Hercynian tectonic reactivation of the Great Glen Fault (e.g. Storetvedt 1987; Storetvedt *et al.* 1990). It is likely that the **A**-magnetization corresponds to the well-established reverse Kiaman superchron of late Carboniferous to late Permian age.

It has long been known that the Anglo–Welsh Curvette has a significant 'Permian' overprinting (Chamalaun & Creer 1964), associated with the Hercynian tectonic deformation of that region. Tectonically, the Dingle Peninsula lies in a similar position as the Anglo–Welsh Cuvette, i.e. on the northern front of the Hercynian orogenic belt. It is not surprising, therefore, that the **A** magnetization of the Dingle Group is well developed, being equally prevalent as the older **B** remanence (*cf.* Fig. 6).

The unfolding test for the two Dingle Group magnetization is outlined in Fig. 7. As seen from Fig. 7 the two *in situ* site mean populations are very well clustered, except for one 'anomalous case' in Group **B** (site 7, with one direction only). However, the progressive increase in directional dispersion, upon structural unfolding, for both magnetizations, implies post-tectonic origin.

The main tectonic disturbances in the Dingle Peninsula seem to pre-date Middle to Upper Old Red Sandstone, as can be demonstrated at Bull's Head by the striking unconformity at which the gently dipping Middle to Upper Devonian Glengarriff Harbour Group rests discordantly upon the strongly disturbed steeply dipping, older strata of the Dingle Group. Therefore, the age of the magnetizations is Middle Devonian or younger.

Owing to the long-lasting problem of finding a positive fold test for any case of **B** magnetization, its Caledonian authenticity has been questioned altogether (Torsvik *et al.* 1989). However, reinterpretation by Deutsch *et al.* (1993) of palaeomagnetic data from the Upper Silurian red siltstone/sandstone succession of Clare Island, western Ireland (Smethurst & Briden 1988) leads to an alternative explanation.

We propose that the **B** field existed for a period of almost 100 Myr, from uppermost Silurian to well into the Middle Carboniferous. Evidence for the older end of the span may be noted in Table 3 where the Dingle Group *in situ* **B** magnetization is compared with corresponding remanence directions from a few other recent studies of rocks in the Irish Caledonides. There is close agreement between the results. The two Clare Island magnetizations, the pre-tectonic **H** (high temperature) remanence and the post-tectonic **I** component are of particular importance in

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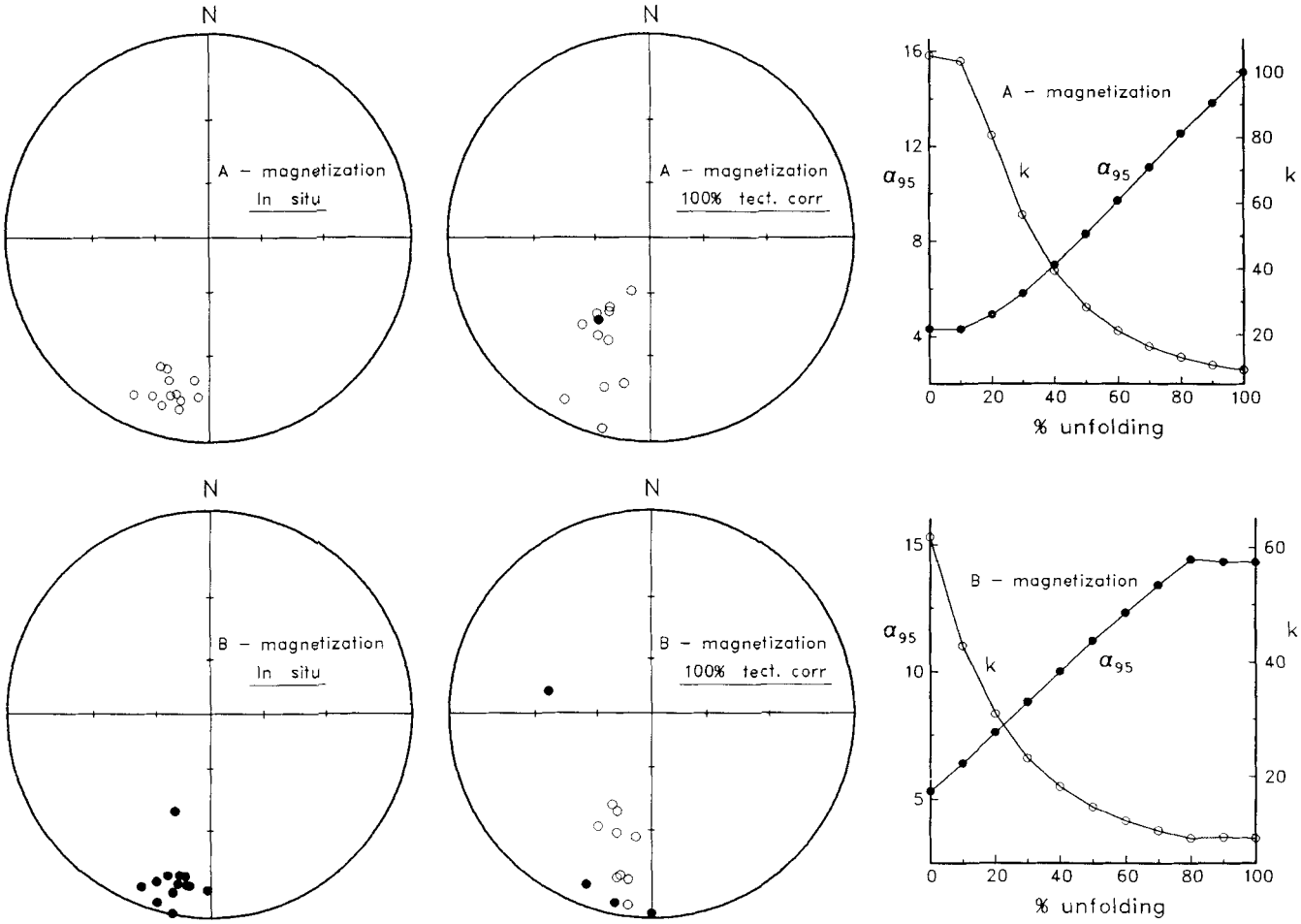



**Figure 7.** Unfolding test for the two characteristic magnetizations of the Dingle Group.

**Table 3.** Overall palaeomagnetic data (**A** and **B** magnetizations for the Dingle Group in comparison with corresponding magnetizations from other rocks in Western Ireland.

| Comp. | Rock Unit | N(s) | D | I | K | $\alpha_{95}$ | Pole | Reference |
|---|---|---|---|---|---|---|---|---|
| A | Dingle Group | 41(N) | 194.3 | -16.0 | 53 | 3.1 | | present paper |
| A | Dingle Group | 12(s) | 194.4 | -14.5 | 105 | 4.3 | 150E, 44N | present paper |
| A | Caherbla/Glengarriff Harbour Groups | 10(N) | 194 | - 24 | 35 | 8.0 | 152E, 49N | Deutsch et al. 1992 |
| B | Dingle Group | 41(N) | 191.0 | 9.2 | 40 | 3.6 | | present paper |
| B | Dingle Group | 13(s) | 192.4 | 9.7 | 62 | 5.3 | 155E, 32N | present paper |
| B | Caherbla/Glengarriff Harbour Groups | 10(s) | 190 | 14 | 41 | 7.6 | 157E, 30N | Deutsch et al. 1992 |
| B | Tourmakeady porphy | 29(N) | 182 | 6 | 51 | 4.0 | 169E, 31N | Deutsch & Storetvedt 1988 |
| B | Salrock I | 16(s) | 195.1 | 13.6 | 19 | 8.8 | 153E, 28N | Smethurst & Briden 1988 |
| $B_{T.C.}$ | Clare Island H | 9(s) | 014.0 | -14.2 | 27 | 10 | 154E, 28N | Smethurst & Briden 1988 |
| B | Clare Island I | 9(s) | 195.5 | -0.5 | 22 | 11 | 151E, 35N | Smethurst & Briden 1988 |

$s$ = number of site mean vectors. Other symbols are as for Table 2. $H$ and $I$ refer to high and intermediate temperature magnetizations as defined by Smethurst & Briden (1988). $B_{T.C.}$ refers to the tectonically corrected Clare Island $H$ magnetization. All other entries are *in situ* remanences. Component notations **A** and **B** are as defined in the present study.

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that the two magnetization axes are not significantly different at the 95 per cent probability level. There is a convincing positive fold test for the **H** component which has a normal polarity. As the main folding is regarded to be Upper Silurian to Lower Devonian (Dewey 1963), the **B** magnetization may have been in existence already from late Silurian time.

For more than 25 years the Siluro–Devonian palaeomagnetic field relative to the British Isles has been a matter of repeated debate (see Torsvik, Trench & Smethurst 1991; Storetvedt *et al.* 1992 for recent contributions to this discussion). In short, some authors have argued that a NE–SW-striking intermediately steep magnetization, reported principally from igneous rocks, represents the true Siluro–Devonian field (originally referred to as the 'revised Devonian pole'). In that view the more predominant shallow-inclined palaeomagnetic axis found in rocks of 'mid'-Palaeozoic age (corresponding to our **B**-magnetization) has been assigned to Hercynian remagnetization. However, the convincing positive fold test for the high-temperature Clare Island magnetization is a strong case against the latter interpretation. Also, recent palaeomagnetic and radiometric studies of rocks from the Cheviot Complex (Mitchell *et al.* 1993; Storetvedt *et al.* 1992) have failed to confirm earlier palaeomagnetic results (Thorning 1974) which have been advanced in favour of the 'revised Devonian pole'. Instead, we have unfolded another case of **B** and **A** magnetizations. The Cheviot Complex has suffered chemical and mineralogical modifications, and all of the rock units have apparently undergone extensive low-temperature oxidation. Other problems with the 'revised Devonian axis' have been discussed elsewhere (Storetvedt *et al.* 1990; Rother & Storetvedt 1991; Storetvedt, Deutsch & Murthy 1991). Therefore, there are reasons to believe that the alleged 'true' Devonian magnetization is an artefact linked to remagnetization processes, and the corresponding data are consequently discounted from the present discussion.

In the Dingle Peninsula, two important tectonic discordances post-date the deposition of the Dingle Group; the first occurs between the Dingle and Caherbla groups (Horne 1974) and is presumably of Lower Devonian age, while the second discordance, between the Caherbla and Glengarriff Harbour groups (Horne 1974) may be assigned to the 'mid'-Devonian. The latter structural discordance may be time equivalent to the one preceding the Upper Old Red Sandstone of Hoy, Orkney, for which Halliday, McAlpine & Mitchell (1977), using the whole rock $^{40}Ar/^{39}Ar$ dating method on associated volcanics, arrived at an age of approximately 370 Ma. From this it follows that the **B** magnetization of the Dingle Group may have been acquired at any time between about 370 and 300 Ma. Owing to the fact that the Hercynian overprinting (evidenced by the **A** magnetization) is relatively strong in the Dingle Group, and that this diastrophic event spans a considerable time interval, beginning in the Lower Carboniferous and continuing well into the Permian, at least a fraction of the **B** population may in principle be as young as Lower to Middle Carboniferous. It is likely that the strength of the remagnetization processes is related to the intensity of associated tectonic deformation (releasing the remagnetizing thermochemical agents). Thus, the fact that the Caledonian movements in the Dingle Peninsula were much stronger than the Hercynian ones, and that the **B** magnetization always occupies the higher blocking temperature range in specimens where the **A** and **B** magnetizations coexist, may indicate that a significant part of the **B** remanence was impressed immediately after termination of the Caledonian deformations ('mid'-Devonian). Another factor in support of this argument is that the main Hercynian orogenic movements in Dingle occurred close to the end of the Carboniferous (Horne 1976), i.e. within the time range of the younger **A** magnetization.

Further clues relevant to the younger end of the proposed age span for the **B** magnetization may be noted from Table 3 (the Salrock **I** and Clare Island **I** magnetizations considered by the original authors as spanning Middle Devonian to Lower Carboniferous ages), and also from results of Lower Carboniferous volcanics of south Scotland (Rother & Storetvedt 1990) and further from the results of the Great Whin Sill (≈310 Ma; Middle Carboniferous), northern England (Storetvedt & Gidskehaug 1969). The authors of the latter study recognized a dominant magnetization similar to what we now call the '**A**' magnetization and discarded a recognizable magnetization (their Fig. 4, group II) similar to what we now call the '**B**' magnetization. If this **B** magnetization in the Great Whin Sill were to be confirmed by future studies, it would provide a younger age limit to the **B** magnetization. The presence of the **B** magnetization in Middle Carboniferous rocks in the British Isles is now well documented in the palaeomagnetic literature and this permits us to propose a wide age range for the **B** magnetization from Upper Silurian to Middle Carboniferous times.

The pre-tectonic origin of the normal polarity Clare Island **H** magnetization may be of importance for interpreting the multicomponent remanence structure of other Siluro–Devonian rocks in the British and Irish Caledonides. It has been found repeatedly (e.g. Storetvedt *et al.* 1990; Deutsch *et al.* 1993) that, in addition to the prevailing reverse post-tectonic **B** magnetization, there is a weak high-temperature magnetization of normal polarity. However, due to experimental noise at higher temperatures, the direction of the weak normal magnetization of the Dingle Group (present study) remains undecided. However, if the high-temperature normal magnetization represents the **B** axis, then a possible magnetization history of the Dingle area may be as below.

(1) An original normal polarity **B** magnetization of Upper Silurian age was subsequently almost completely erased by Acadian and Hercynian thermochemical processes.

(2) The reverse polarity, ubiquitous, **B** magnetization was probably acquired mainly in 'mid'-Devonian time, immediately following the Caledonian diastrophic events. However, part of this magnetization might have been impressed by Hercynian events, extending well into Middle Carboniferous times.

(3) The reverse polarity **A** magnetization, equally as prominent as the reverse **B** magnetization, was acquired by thermochemical agents released by the Hercynian orogeny posterior to the onset of the Kiaman superchron.

In the above-suggested history, the earliest phase (1) is the most speculative regarding the magnetization age and indeed its remanence direction. However, the presence of

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the normal pretectonic **H** component for Clare Island allows us to suggest that a late Silurian to early Devonian normal polarity high-temperature component has marginally survived in the Dingle Group rocks.

In summary, results from the Dingle Group presented above provide a magnetization history broadly conforming to the pattern reported from other localities in the Irish and British Caledonides.

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